L_0 regularization for subnational microsimulation calibration

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Abstract

Subnational microsimulation needs survey microdata that reproduce administrative totals across nested geographies while staying usable inside a policy model. A fidelity-oriented pipeline can combine many sources and add record-level variation, producing a candidate dataset rich enough to represent the targets but too large to simulate. That dataset has to be reduced, and the question is which records to keep.

We study this reduction as a sampling problem. Using L_0 regularization with the Hard Concrete relaxation of Louizos et al. (2018), the method jointly selects which records to retain and assigns each a positive weight, with selection driven by the calibration targets rather than drawn at random. The gates make the retained-record count an explicit optimization variable, and soft and hard concentration controls bound how large individual weights can grow. Because the optimization that selects records also calibrates them, the reduced dataset continues to match its targets.

We implement the method within POPULACE, PolicyEngine's microsimulation data pipeline, and present a proof of concept on its national and state target surface. Holding the candidate universe, the target set, and the record budget fixed, we vary only how records are chosen across four samplers: informed L_0 selection, its convex L_1 analog, random sampling followed by reweighting, and survey-weight (Hansen–Hurwitz) sampling. All fit weights with the same calibrator, so the experiment isolates selection. Under aggressive compression informed L_0 keeps out-of-sample error bounded where a small random draw cannot: at a 2,000-record budget its out-of-sample median absolute relative error is 47.3% against random sampling's 50.1%, and on the tail-sensitive mean the gap is far wider (225% against 1,798%). As the budget grows the median edge passes to random sampling, but informed L_0 keeps the lower mean throughout, avoiding the seed-to-seed instability of a small draw. The convex L_1 arm collapses under forced sparsity—one penalty cannot both select records and keep their weights—and recovers toward L_0 only at large budgets. The informed L_0

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method's distinctive value is operational rather than a uniform accuracy gain: it sets dataset size, geographic emphasis, and weight concentration through explicit parameters in one optimization, and removes the variance of a poor random draw at tight budgets. Matching a demanding target system can still concentrate population mass on a few records, a cost we report rather than hide. We leave the larger build-and-prune and subnational settings the controls are designed for to future work.

Keywords: microsimulation, survey calibration, subsampling, L_0 regularization, subnational analysis

1 Introduction

Tax-benefit microsimulation models apply policy rules to household microdata to estimate fiscal and distributional effects. At the national level, that usually means starting from a survey such as the Current Population Survey (CPS), improving measured incomes and program participation, and adjusting weights so the dataset reproduces known aggregates. Subnational work is harder. Analysts often want estimates for states, congressional districts, or local areas, but the underlying survey was not designed to support all of those geographies at once.

That gap matters in practice, because many reforms are proposed, administered, or debated below the national level. A state tax proposal needs state-specific incomes, filing patterns, and benefit participation. District-level analysis needs those same outcomes to aggregate correctly within each district while staying consistent with state and national totals. A usable subnational dataset therefore has to satisfy a hierarchical calibration problem rather than a single national one.

One way to meet that demand is to build a dataset that carries as much of the relevant variation as possible. A pipeline can combine several surveys and administrative files, impute the variables that any single source lacks, and attach the geographic detail that local analysis needs. Each of these steps enriches the records, and some add records. The result is a candidate dataset rich enough to represent the variation the targets describe, and large enough that it becomes expensive to store, calibrate, and simulate. A file with millions of active records supports detailed subnational work but is slow to run; a much smaller file is easier to ship, but only if the loss in fidelity is acceptable.

The candidate dataset therefore has to be reduced to a size that can be shipped and simulated, and this paper treats that reduction as a sampling problem: given the large candidate universe and a fixed record budget, which records should the final dataset keep, and what weight should each kept record carry? The simplest answer is to draw a random subset and reweight it to the targets. We study an alternative in which the selection itself is informed by the targets.

The method is L_0 regularization with the Hard Concrete relaxation (Louizos et al., 2018). Each candidate record receives a gate that the optimizer can drive toward zero or one, and the expected number of open gates enters the objective directly. Minimizing calibration error and the gate count together selects a subset of records and fits their weights in one optimization. The retained-record count becomes a quantity the optimizer controls rather than a fixed draw, and two concentration controls (a soft penalty and a hard per-record cap) limit how concentrated the fitted weights can become. In practical terms, one framework can target a compact national dataset or a larger subnational dataset, at a chosen record budget, with weights kept inside a usable range.

We situate this work in POPULACE (PolicyEngine, 2026b), PolicyEngine's pipeline for microsimulation dataset construction. POPULACE passes a single weighted data structure through a sequence of modular stages—source loading, imputation, geographic assignment, target construction, and calibration—each implemented as a separate package operating on that shared structure. Because the stages are modular, the data sources, the imputation model, the geographic assignment, and the calibration method are interchangeable components rather than fixed code, and the calibration step studied here is one such component. The paper describes the pipeline at the level needed to follow the calibration step, and then describes the configuration used here.

This paper is a proof of concept on POPULACE's current target surface: it asks whether L_0 selection with Hard Concrete gates can produce a sparse, calibrated dataset on the targets POPULACE assembles today. Local-area and subnational calibration are the production motivation for the method and a future application of it, rather than results this paper presents.

The empirical question is whether informed selection earns its added complexity. We ask three things. First, when does informed L_0 selection reproduce the targets more accurately than random sampling, survey-weight sampling, and the convex L_1 alternative at the same record budget, and how does that comparison change as the budget tightens? Second, does the retained sample generalize, in the sense that it also matches whole target families held out of the fit? Third, does the framework give usable control over the size of the output, the geographic level it focuses on, and the concentration of the fitted weights, and what does tightening the concentration control cost in accuracy? The answers are regime-dependent: informed selection's accuracy advantage over the baselines concentrates at small budgets, while its control over size, geography, and weight concentration is what distinguishes it at the larger budgets at which a dataset is shipped.

The rest of the paper follows that structure. Section 2 places the work in the literature on spatial microsimulation, survey calibration, and sparse selection. Section 3 describes the base microdata, the POPULACE pipeline, and the administrative targets. Section 4 presents the calibration method and the sampling experiment design. Section 5 reports the comparison against the sampling baselines and the behaviour of the size, geography, and weight controls. Section 6 interprets those results and discusses their limits, and Section 8 concludes.

2 Background and related work

2.1 Subnational microsimulation and spatial reweighting

Subnational microsimulation starts from a recurring problem: a national survey carries the behavioural and demographic detail needed for policy modelling, but it does not contain enough observations in every local area to support direct estimation. Spatial microsimulation addresses that gap either by reweighting existing survey records or by building synthetic small-area populations from aggregate constraints (Tanton, 2014; O'Donoghue et al., 2014).

This distinction matters for the present work, because our pipeline is a reweighting system built on observed CPS households. It places households across geographies to create local variants, but the core problem remains the calibration of survey-based microdata to administrative targets, which keeps calibration methods as the closest reference point.

The spatial microsimulation literature supplies further context. Williamson et al. (1998), Huang and Williamson (2001), and Harland et al. (2012) describe combinatorial and synthetic-population approaches that search over record assignments area by area. Tanton et al. (2011) and Lovelace and Dumont (2016) discuss reweighting for small-area estimation and policy analysis. These methods usually target one geographic layer at a time, whereas our setting asks one weighted dataset to reproduce state and national targets together.

2.2 Survey calibration

Let $i = 1, \ldots, n$ index sampled units, let d_i denote the initial design weight for unit i, and let $\mathbf{x}_i \in \mathbb{R}^p$ denote the auxiliary variables used for calibration. Survey calibration seeks adjusted weights w_i whose weighted totals match known population totals:

$$\sum_{i=1}^{n} w_i \mathbf{x}_i = \mathbf{T}, \tag{1}$$

where $\mathbf{T} \in \mathbb{R}^p$ is the vector of published totals (Deville and Särndal, 1992; Särndal, 2007). The auxiliary variables may be counts, indicators, or continuous quantities such as income components.

The subnational problem extends this setup in two ways. The target vector draws on several administrative sources, so it mixes count and dollar constraints. The constraints are also nested by geography: district totals should sum to state totals, which should sum to national totals after uprating and reconciliation. At production scale this yields a large target system with substantial collinearity across rows.

2.3 Calibration estimators

Two classical methods are the standard references for this kind of problem. Generalized regression (GREG) minimizes a distance from the initial weights subject to the calibration equations (Deville and Särndal, 1992; Särndal, 2007). It accepts quantitative targets directly and is well understood, but it can return negative weights and becomes harder to solve as the target system grows and its rows become collinear. Negative weights are acceptable in some estimation settings and awkward in a microdata file meant for tax-benefit simulation, where a household is expected to represent positive population mass.

Iterative proportional fitting (IPF, or raking; Deming and Stephan, 1940; Ireland and Kullback, 1968) adjusts weights multiplicatively so that selected margins match published totals. It preserves non-negativity and avoids matrix inversion, which makes it a natural choice for count-style margins. In spatial microsimulation and population synthesis, raking is commonly paired with a sampling step in one of two orders: draw a sample first and rake that fixed sample, or rake the full file first and integerise the fractional weights into a finite population. Iterative proportional updating and related extensions handle household and person margins simultaneously (Pritchard and Miller, 2012). These raking-based workflows are natural comparators for pruning methods on categorical margin surfaces. They are not direct full-surface baselines here: classical IPF is built for non-negative categorical margins, whereas the production target surface mixes counts, dollar totals, overlapping families, near-zero targets, and hierarchical constraints. Generalized raking or entropy calibration can extend the idea to broader linear constraints, but then the method becomes a general calibration optimizer rather than the classical IPF workflow.

A third approach treats calibration as an optimization problem to be solved numerically. The two methods above are special cases of one problem: Deville and Särndal (1992) define the calibrated weights as those closest to the design weights, under a chosen distance, subject to reproducing the targets, with GREG and raking the closed-form solutions for particular distances. When the target system is large, collinear, mixes count and dollar constraints, and restricts weights to be positive, the classical Newton and Lagrange solvers become hard to apply or fail to converge, which motivates minimizing the calibration objective directly. Espuny-Pujol et al. (2018) do so with a global optimization that enforces the range restrictions on the weights explicitly, and minimum-distance reweighting has long been used this way in tax microsimulation (Creedy, 2003). At the scale of an enhanced microdata file the practical solver is stochastic gradient descent: the weights are parameterized to stay positive, typically in log space; a loss measures the relative discrepancy between the weighted estimates and the targets; and an optimizer such as Adam (Kingma and Ba, 2015) minimizes it, which scales to thousands of mixed, hierarchical targets without inverting a matrix. Woodruff and Ghenis (2024) calibrate PolicyEngine's enhanced CPS in exactly this way.

These estimators answer a calibration question rather than a sampling one: given a fixed set of records, they find weights that match the targets. The method studied here selects records and fits weights at the same time, extending the gradient-based calibration just described with explicit record selection. Classical calibration still matters for the comparison set, because it can be combined with selection: sampling followed by raking is the classical reduce-then-calibrate analogue to random sampling followed by gradient reweighting, while raking followed by integerisation is the classical calibrate-then-reduce analogue to sampling from dense calibrated weights.

2.4 Reducing microdata by selecting records

Reducing a large candidate dataset to a fixed budget is a sampling problem with a long statistical history, and several established methods reduce a microdata file by selecting actual records to match known totals. They differ mainly in where calibration enters: before selection, after selection, or inside the selection step itself. We review the broad option space because it defines what a reasonable analyst might try: random sampling followed by calibration, calibration followed by integerisation, raking-based variants of both, balanced sampling, combinatorial optimisation, and sparse optimization relaxations. Not all of these are equally applicable to the production problem. The target surface here is large, mixed, and hierarchical, so methods built for categorical margins or discrete area-by-area searches are reference points rather than direct full-surface baselines. The matched-budget comparison in Section 4 therefore focuses on the tractable full-surface set: informed L_0, its convex L_1 analog, random sampling with reweighting, and survey-weight sampling. Raking-based reductions, balanced sampling, and combinatorial optimisation remain published comparators that motivate robustness checks where their assumptions fit cleanly. We also note the coreset view from machine learning, where a weighted subset is chosen so that a model fit on the subset approximates the fit on the full data (Feldman, 2020; Bachem et al., 2017).

2.4.1 Random sampling

The simplest reduction draws records at random. Under simple random sampling without replacement each record has inclusion probability n/N and design weight N/n, and the Horvitz–Thompson estimator recovers population totals without bias (Horvitz and Thompson, 1952; Cochran, 1977; Särndal et al., 1992). A random subsample preserves the conditional distribution of the data in expectation, but it does not reproduce administrative totals, so it has to be calibrated after the draw. Random selection is also the reference that the data-reduction literature treats as the genuine test: informativeness-based schemes are measured against it (Ma et al., 2015), and pruning methods in deep learning often fail to beat it at high reduction ratios (Guo et al., 2022). In our setting it is the reduce-first, calibrate-after baseline, a random draw followed by gradient-descent reweighting to the targets. Replacing the gradient reweighting step with raking gives the classical sampling-then-raking workflow; it is directly comparable on a raking-compatible categorical margin subset, but not on the full mixed count-and-dollar target surface without changing the problem into generalized calibration.

2.4.2 Raking, survey-weight sampling, and integerisation

A second approach calibrates first and reduces second. Given fractional weights from a calibration step, survey-weight sampling selects records with probability proportional to those weights, which turns a reweighted file into a finite, integer-weighted population that approximately preserves both the calibrated totals and the distribution. This is unequal-probability, probability-proportional-to-size sampling in the survey tradition (Hansen and Hurwitz, 1943; Horvitz and Thompson, 1952), applied to microsimulation as integerisation. The upstream calibration can be IPF/raking, GREG, or a numerical optimizer; the downstream question is how to convert fractional weights into a finite file. Lovelace and Ballas (2013) catalogue the common integerisation methods and show that truncate-replicate-sample reproduces the targets most accurately while fixing the reduced population to an exact size; earlier top-up schemes are due to Ballas et al. (2005). Because the selection probabilities are the calibration weights, survey-weight sampling presupposes a calibration step, so in our comparison it is the calibrate-then-reduce baseline, with gradient descent supplying the weights. A raking-then-TRS variant is the corresponding classical baseline on a categorical margin surface, but it does not naturally extend to the full mixed production target surface without replacing IPF with a broader calibration solver.

2.4.3 Balanced sampling

Balanced sampling selects a fixed-size sample whose Horvitz–Thompson estimates reproduce auxiliary totals exactly or nearly exactly. The cube method of Deville and Tillé (2004) is the standard construction: it chooses inclusion indicators while preserving prescribed balancing equations, then uses the resulting sample weights for estimation. This makes balanced sampling a target-aware selection method rather than a post-sampling calibration method, and therefore a relevant published neighbour of L_0 selection. Its fit to the present problem is not exact. Classical balanced sampling usually treats the auxiliary totals as design constraints with predetermined inclusion probabilities, whereas the production calibration surface here mixes categorical counts, dollar totals, and hierarchical relationships and then fits positive weights after selection. Still, it is an important reference point for the idea that the sample itself, not only its weights, can be chosen to preserve known totals.

2.4.4 Convex sparse calibration

A final family keeps the continuous optimization setup but replaces the direct retained-record count with a convex sparsity surrogate. An L_1 penalty on fitted weights, implemented with proximal soft-thresholding, can drive weights exactly to zero while preserving the same calibration loss. This is tractable on the full target surface and gives a useful ablation: it asks whether the value comes from sparse joint selection in general or from the Hard Concrete L_0 relaxation specifically. Its drawback is interpretive. The L_1 penalty shrinks weight magnitudes as well as encouraging zeros, so its tuning parameter is not a direct retained-record-count control in the way the expected L_0 norm is. We therefore treat it as a convex sparse analogue rather than as the primary sampling method.

2.4.5 Informed selection

These methods share the idea that a carefully chosen weighted subset can stand in for a larger dataset, but they reach it differently: random sampling ignores the targets and repairs the gap afterwards, survey-weight sampling inherits the targets from a prior calibration, balanced sampling tries to preserve auxiliary totals through the sample design, and convex sparse calibration remains in the same optimizer but changes the penalty. A fourth tradition makes selection itself the calibration: combinatorial optimisation chooses a combination of records, with integer multiplicities, to match the benchmark totals directly, searched with metaheuristics such as simulated annealing (Kirkpatrick et al., 1983; Williamson et al., 1998; Voas and Williamson, 2000). It is the closest precedent for target-informed selection, but it is discrete and gradient-free; we note it as related work rather than include it in the matched-budget comparison. The method studied here also pursues the targets at selection, but with a continuous, gradient-based relaxation that fits the selection and the weights jointly. The next subsection sets out that relaxation.

2.5 L_0 regularization and the Hard Concrete distribution

The L_0 norm counts the non-zero entries in a vector. Applied to record weights, it counts the records that remain active, which is exactly the size of the retained sample. Selecting a subset of a chosen size is therefore an L_0-penalized problem. The difficulty is that the L_0 norm is non-differentiable and combinatorial, so it cannot be minimized by gradient descent in its raw form.

Louizos et al. (2018) make the penalty differentiable by attaching a stochastic gate $z_i \in [0, 1]$ to each parameter and optimizing the expected L_0 norm. The gate is drawn from a Hard Concrete distribution: a binary concrete (continuous-relaxation) variable stretched beyond the unit interval and then clipped back into it. With a learned location parameter $\log \alpha_i$ and a fixed temperature β, draw $u \sim \mathrm{Uniform}(0, 1)$ and set

$$s_i = \sigma\left(\frac{\log u - \log(1-u) + \log \alpha_i}{\beta}\right), \tag{2}$$

$$\bar{s}_i = s_i(\zeta - \gamma) + \gamma, \tag{3}$$

$$z_i = \min\big(1, \max(0, \bar{s}_i)\big), \tag{4}$$

where σ is the logistic function and $\gamma < 0 < 1 < \zeta$ are fixed stretch bounds. Stretching s_i from $(0, 1)$ to (γ, ζ) and clipping to $[0, 1]$ places point masses at exactly 0 and exactly 1, so a gate can switch a record fully off or fully on while staying differentiable in between.

Because the gate has a closed-form probability of being non-zero, the expected L_0 norm is differentiable. The probability that gate i is active is

$$\Pr(z_i \neq 0) = \sigma\left(\log \alpha_i - \beta \log \frac{-\gamma}{\zeta}\right), \tag{5}$$

and the L_0 penalty is the sum of these probabilities over all records, $\sum_i \Pr(z_i \neq 0)$, which is the expected number of retained records; minimizing it pushes gates toward zero. At test time the noise is removed and the gate is evaluated deterministically,

$$\hat{z}_i = \min(1, \max(0,\, \sigma(\log \alpha_i)(\zeta - \gamma) + \gamma))\,, \tag{6}$$

which returns an exact zero for any record driven far enough off and an exact one for any record driven far enough on, with intermediate values in between. The exact zeros make the weight vector exactly sparse; a record left with an intermediate gate is retained with its fitted weight scaled by the gate.

The two pieces depend on each other, which is the point worth stressing for readers new to this construction. The L_0 penalty states the objective, that fewer active records are preferred, but on its own it is not optimizable by gradient methods. The Hard Concrete gate makes that objective differentiable, but on its own, without the penalty, it carries no pressure toward sparsity and leaves every gate open. Used together they turn record selection into a term the optimizer can minimize alongside calibration error. This is what makes the retained-record count a control variable rather than the result of a separate thresholding step.

2.6 Convex sparse selection (L_1)

The L_0 count is the natural sparsity objective but a hard one. The standard convex surrogate replaces it with the L_1 norm of the weights, $\|\mathbf{w}\|_1 = \sum_i |w_i|$, which is the LASSO of Tibshirani (1996). Adding the penalty to the calibration loss gives the convex program

$$\min_{\mathbf{w}\geq 0}\ \mathcal{L}_{\text{cal}}(\mathbf{w}) + \lambda_{L_1} \sum_{i=1}^{n} w_i, \tag{7}$$

where λ_{L_1} sets the strength of the penalty and the weights are non-negative here, so $|w_i| = w_i$. The L_1 norm is the standard tool for sparse regression and feature selection: unlike the smooth L_2 (ridge) penalty, which shrinks coefficients but rarely sets them to zero, its corner at the origin drives small coefficients to exact zeros. The program is convex and is minimized by proximal gradient descent, each gradient step followed by a soft-thresholding step $w_i \leftarrow \max(w_i - \tau, 0)$ that subtracts a fixed amount $\tau \propto \lambda_{L_1}$ from every weight and clips at zero (Beck and Teboulle, 2009), so a record whose pull on the targets does not clear the threshold is dropped while the rest are kept and shrunk. The trade-off is structural: the single penalty λ_{L_1} controls both how many records survive and how much their weights shrink, because the same norm that zeroes small weights also pulls the surviving ones toward zero, so forcing a small sample forces heavy shrinkage on the records that remain. We include this convex arm as the direct comparison to L_0: it pursues the same target-informed sparse selection by a more stable route, and contrasting the two isolates what the L_0 gates buy by separating selection from weighting.

2.7 Microsimulation dataset engines

Beyond the calibration step itself, Woodruff and Ghenis (2024) embed that gradient-based reweighting in a national pipeline that first imputes tax variables from the IRS Public Use File onto CPS records, and that regularizes the weights with dropout rather than an explicit selection mechanism. The present work keeps the gradient-based calibration but replaces dropout with the L_0 and Hard Concrete gates of the previous subsection, so that record selection becomes explicit and the fitted weights are exactly sparse. It also runs within POPULACE, a pipeline that expresses dataset construction as a sequence of modular stages over a shared data structure. The data sources, the imputation model, the geographic assignment, and the calibration method are interchangeable components rather than fixed steps, which lets the same pipeline support different countries and different methodological choices. Section 3 describes the pipeline and the configuration used here.

3 Data

The calibration problem is defined by two inputs: the household microdata that form the candidate universe, and the administrative targets the calibrated dataset must reproduce. This section describes how POPULACE assembles the candidate microdata and how this study selects its targets. The pipeline is the harness that produces the calibration problem; it is not itself the object of evaluation.

3.1 Populace as a construction pipeline

Populace builds a microsimulation dataset by passing a single weighted data structure—a sampling frame holding the records, their weights, and provenance—through a sequence of stages. Each stage is a separate package that reads and writes that shared frame: it loads the source surveys, combines them into one record universe, imputes missing variables, assigns geography, constructs the calibration targets, and calibrates. Because every stage operates on the same structure, the imputation model and the calibration method are components that can be swapped without rewriting the pipeline, and the method studied here is one calibration option among those the pipeline supports.

This modular structure makes the candidate universe a property of the configuration rather than an accident of the code: the source-loading and imputation stages set what information each record carries, and calibration sets which records survive and at what weight. Populace follows a generate-then-prune design, assembling the candidate universe once and pruning it to a deployable budget at the calibration stage.

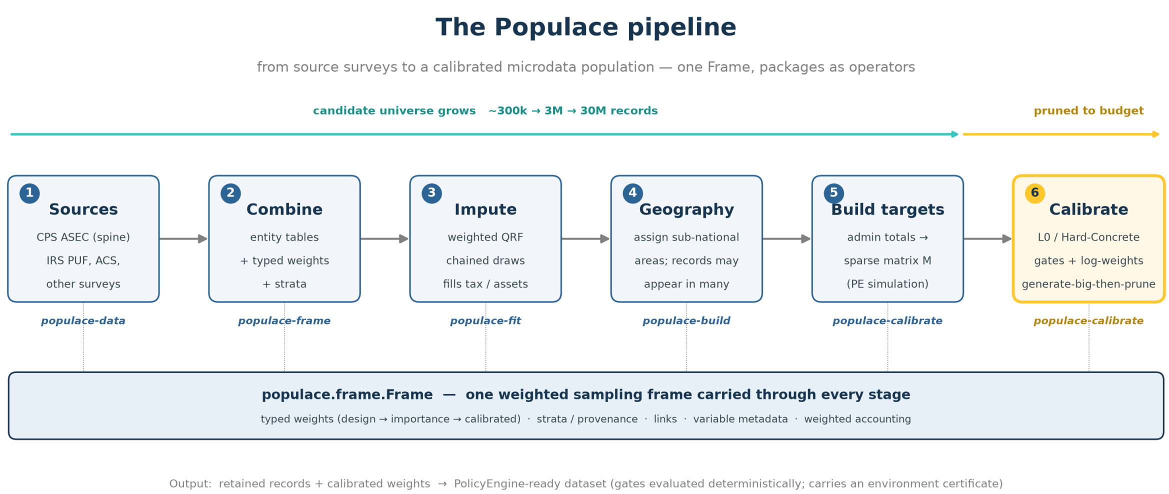

Figure 1: Populace pipeline for the configuration used in this paper: load and combine the source surveys, impute missing variables, assign geography, construct the calibration targets, and calibrate. The calibration stage is the focus of this paper.

For this build the candidate universe is the Populace-US 2024 household file: 75,112 households at period 2024, one row per household. We pin the exact artifact for reproducibility: HuggingFace repository `policyengine/populace-us`, snapshot `be80a14f`, built from POPULACE commit `6e1bcd0`, with H5 SHA-256 beginning `f0af2519`. The repository's `main` branch now points to a newer release, so the numbers reported here are tied to this pinned snapshot and file hash.

3.2 Base microdata

3.2.1 Current Population Survey

The primary source is the Current Population Survey Annual Social and Economic Supplement (CPS ASEC; U.S. Census Bureau, 2024), a nationally representative household survey from the US Census Bureau and the Bureau of Labor Statistics. The CPS provides demographic structure, household relationships, labour income, transfer income, and reported participation in programmes such as SNAP, Medicaid, SSI, and TANF.

The CPS is not a complete tax-benefit dataset on its own. Some tax variables are missing, others are measured with error, and top incomes are imperfectly captured relative to tax records (Burkhauser et al., 2012). Benefit receipt is also underreported relative to administrative sources (Meyer et al., 2015). The pipeline therefore augments the CPS before calibration.

3.2.2 Imputation from administrative and survey sources

The pipeline enriches the CPS base by imputing variables from administrative and survey donors. It imputes tax-return detail from the IRS Public Use File (PUF; Bryant, 2023) onto CPS records following Woodruff and Ghenis (2024): the PUF carries detailed tax-return information but lacks household structure and demographic coverage, so quantile regression forests (Meinshausen, 2006) predict the PUF variables on the CPS; Meyer et al. (2020) assess the accuracy of such survey–administrative tax imputations. The imputation is sequential and regime-gated, drawing from a weighted bootstrap, so later variables condition on earlier imputed values. The same machinery draws further variables from the donors in Table 1 where the CPS is thin. Each step adds variables to the existing households rather than new records, so the candidate universe stays at one row per household.

Donor source	Variables contributed (this build)
CPS ASEC	Household and person structure, demographics, labour and transfer income, programme participation, unit assignment
IRS PUF (2015, uprated)	Itemized-deduction detail, QBI/SSTB components, partnership and S-corp income, selected tax detail
CPS ORG	Hourly wage, paid-hourly and union status, weekly hours, overtime inputs
SCF	Wealth and net-worth components
SIPP	Tips
MEPS-IC	Employer-sponsored insurance premiums
ACS (2022)	Rent (pre-subsidy)
CMS marketplace PUFs and CPS ASEC	ACA take-up and the benchmark-plan ratio

Table 1: Donor sources imputed onto the CPS base for this build, from the POPULACE-US release stage manifest. The stages run in a fixed order, each conditioning on the variables already imputed.

3.2.3 Geography

Each household carries a geographic identity. The CPS ASEC reports the household's state, which supports the state-level administrative targets this study calibrates to. The assembled file also carries county, congressional-district, tract, and block identifiers, but this study calibrates only to national and state targets, so the finer levels are present without being scored. For this build the candidate universe is one row per household: records are not replicated across alternative geographic placements, so the candidate count is the household count of 75,112.

3.3 Calibration targets

The target system combines aggregates from several administrative sources at the national and state levels. This study does not calibrate to every target the pipeline could construct. It calibrates to a prioritized subset, and it reports that subset explicitly rather than describing it as a complete or maximal target set. Appendix A lists the target families used and the reason for including them. The exact counts come from each experiment's run manifest rather than from a pipeline-wide inventory.

Target family	Geographic level	Targets
IRS Statistics of Income (irs_soi)	National, state	3,107
Census population estimates (census_population)	National, state	936
Medicaid and CHIP, CMS (cms_medicaid)	National, state	104
ACA marketplace, CMS (cms_aca)	State	102
SNAP, USDA (usda_snap)	National, state	52
State income tax, Census STC (state_income_tax)	State	45
TANF, HHS ACF (hhs_acf_tanf)	National, state	30
Social Security OASDI, SSA (ssa)	National	6
CBO revenue projections (cbo)	National	5
JCT tax expenditures (jct)	National	5
Medicare, CMS (cms_medicare)	National	1
Total		**4,393**

Table 2: Prioritized target subset used in this study, by family and geographic level. The subset is chosen for the reasons given in Appendix A and is not a complete inventory of the targets the pipeline can construct. All targets are at the national or state level; the surface holds no congressional-district targets.

Counts are read from the run manifest. The cbo family is held out as a validation-only diagnostic and is never used to fit weights (Section 4.5); the source for each family is listed in Appendix A.

4 Methodology

This section describes the calibration method that selects and weights records, and then the design of the experiment that evaluates the method as a sampler.

4.1 The shared calibrator

The calibration matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$ maps n candidate records to m targets. Entry M_{ji} is the contribution of candidate i to target j: an indicator or count for count targets, and a simulated tax or benefit value for dollar targets. The matrix is built from POLICYENGINE simulations evaluated under the policy rules that apply to each record's assigned geography, and is stored in sparse form. Given a weight vector $\mathbf{w} \in \mathbb{R}^n$, the estimate of target j is

$$\hat{t}_j = \sum_{i=1}^{n} M_{ji}\, w_i. \tag{8}$$

The four methods compared in this study fit weights with the same calibrator, the `calibrate` routine in POPULACE. It minimizes a capped, weighted mean absolute percentage error (capped weighted MAPE) of the estimates against the targets,

$$\mathcal{L}_{\mathrm{cal}}(\mathbf{w}) = \frac{\sum_{j=1}^{m} \omega_j \min\left(\left| \frac{\hat{t}_j - t_j}{s_j} \right|, c \right)}{\sum_{j=1}^{m} \omega_j}, \tag{9}$$

where $s_j = \max(|t_j|, 1)$ scales each target by its own magnitude, with a floor of one unit so a zero-valued target stays well defined; c caps the contribution of any single hard-to-fit target, so one badly conditioned target cannot dominate the gradient, and is set to the production value $c = 1$ in the runs reported here (Appendix B contrasts the looser $c = 10$); and ω_j are per-target weights, normalised by their sum so only their relative magnitudes matter. The error is relative, so a one per cent miss on a small target and on a large one count the same. The weights are parameterized in log space, $w_i = \exp(\theta_i)$, so the fitted weights stay positive; an Adam optimizer (Kingma and Ba, 2015) implemented in PyTorch (Paszke et al., 2019) minimizes the loss, and the total weight is held to the input population. On a fixed set of records this calibrator answers a calibration question: it finds weights that match the targets. The random and survey-weight baselines use it unchanged; informed L_0 and informed L_1 extend it with a selection term.

The per-target weights ω_j in the runs reported here follow POPULACE's production US-fiscal weighting, computed through the same code path the production US build uses rather than a separate reimplementation. The weighting is built in four steps. Each target is first assigned to one of two value bases from its measure metadata: a count basis (indicator, enrollment, recipient, and return-count measures) or an amount basis (dollar measures). Within a basis, a target's raw weight is proportional to $\max(|t_j|, 1)^{1/2}$, the square root of its own magnitude, so a larger aggregate carries more weight but sublinearly (a target a hundred times larger receives about ten times the weight), and these raw weights are normalised to mean one inside each basis. The two bases are then rescaled to contribute equal total weight, so the far more numerous dollar cells do not swamp the count targets. A final step sets the mean weight to one. Because the loss divides by $\sum_j \omega_j$, only the relative weights enter the objective and their absolute scale does not.

4.2 Selection and weighting as one optimization in our proposed method

Informed L_0 sampling adds two terms to the shared calibrator that turn weight fitting into record selection. Each candidate i carries a Hard Concrete gate $z_i \in [0,1]$ (Louizos et al., 2018), the gated estimate becomes $\hat{t}_j = \sum_i M_{ji} w_i z_i$, and the optimizer minimizes

$$\mathcal{L}(\mathbf{w}, \boldsymbol{\alpha}) = \mathcal{L}_{\text{cal}}(\mathbf{w}, \mathbf{z}) + \lambda_{L_0} \sum_{i=1}^{n} \bar{z}_i + \lambda_{L_2} \frac{1}{n} \sum_{i=1}^{n} \left(\frac{w_i}{w_{0,i}} \right)^2 . \tag{10}$$

The first term is the capped weighted MAPE of Equation 9, now evaluated on the gated weights $w_i z_i$. The second term is the L_0 penalty and sets the sample size: $\bar{z}_i = \Pr(z_i \neq 0)$ is the expected activation of gate i (Equation 5), so $\sum_i \bar{z}_i$ is the expected number of open gates, the expected retained-record count, and λ_{L_0} sets how aggressively the optimizer prunes. Rather than set λ_{L_0} by hand, each build tunes it by an outer bisection to a requested record count (Appendix C). The third term is a soft concentration penalty: λ_{L_2} multiplies the mean squared ratio of each fitted weight to its initial weight $w_{0,i}$, which discourages the optimizer from carrying the fit on a few heavily inflated records. It applies to informed L_0 only; the random and survey-weight baselines run the shared calibrator without any selection or concentration term.

Read as a sampler, the calibration term keeps the retained sample faithful to the targets, the L_0 penalty sets the sample size, and the L_2 penalty governs weight concentration. The concentration term matters because an L_0 penalty on its own can reach a sparse solution that is still unusable: the optimizer can match the targets by placing very large weights on a few retained records. Both concentration controls act on the ratio of each fitted weight to its initial weight, $w_i/w_{0,i}$. The L_2 term is a soft penalty on the mean square of that ratio that discourages large inflations smoothly, and it is the control we sweep to trace the accuracy-concentration trade-off. `max_weight_ratio` is a hard per-record cap on the same ratio: no fitted weight may exceed a fixed multiple of that record's initial weight, clamped after every optimizer step. The soft penalty shapes the whole weight distribution during optimization; the cap bounds the single largest weight. Size and concentration are therefore set by separate controls: λ_{L_0} fixes how many records remain, and the L_2 penalty and `max_weight_ratio` fix how concentrated their weights may become. The headline runs leave the cap off and set $\lambda_{L_2} = 0$, so the reported accuracy frontier is the pure-calibration result; Section 5.3 reports $\lambda_{L_2} = 10^{-4}$ beside it as an operability contrast, tracing what the penalty buys in effective sample size against what it costs in fit. Tightening either control bounds how much population a single record can carry, at some cost to calibration fit. Appendix D lists the full configuration, and Appendix E gives the optimization loop as Algorithm 1.

The gates and the weights are learned together: the gate decides whether a record is in the sample, and the weight decides how much it counts. Because both are fit against the same calibration loss, selection is informed by the targets: a record survives when keeping it helps match a target that would otherwise be missed. This is the sense in which the method is a target-informed sampler rather than a reweighting of a fixed random draw.

4.3 From fitted weights to a dataset

After optimization the gates are evaluated deterministically, so the retained-record count is read from the solution rather than chosen by a post-hoc cutoff. The retained records and their weights are assembled into a dataset for POLICYENGINE. Appendix F summarizes the assembly step, which matters operationally but is not the object of the experiment.

4.4 Sampling experiment design

The experiment evaluates whether informed selection produces a better dataset than the alternative samplers at the same record budget. The literature offers more possible comparators than can be run cleanly on the full mixed target surface: raking-based workflows are natural on categorical margins, balanced sampling is designed for fixed balancing variables, and combinatorial optimisation is discrete and costly at production scale. The headline comparison therefore keeps the candidate universe, target surface, positivity restriction, and calibration loss fixed, and varies only the way records are selected. From one candidate universe and one target set, we compare four samplers at a matched record budget: informed L_0 selection, its convex L_1 analog, and two classical reduction baselines that bracket where calibration enters (Section 2.4):

- **Informed L_0 sampling with Hard Concrete gates.** The shared calibrator with the gate and concentration terms above selects records and fits their weights jointly at a chosen budget, set through λ_{L_0}.
- **Informed L_1 selection.** The same shared calibrator with the Hard Concrete L_0 term replaced by a convex L_1 penalty on the weights, solved by a proximal (iterative soft-thresholding) step that drives unneeded weights to exact zeros (Section 2.6). As with informed L_0, an outer bisection tunes the penalty λ_{L_1} to the requested record count. It is the joint analog of informed L_0—selection and weighting share one objective

with no post-selection reweighting—so a single penalty must serve both, which is the property the comparison isolates.

- **Random sampling with gradient-descent reweighting.** A random subset of the candidate universe of the same size is drawn, then the shared calibrator fits its weights to the same targets without the gate or concentration terms. This is the reduce-first, calibrate-after baseline.

- **Survey-weight sampling (Hansen–Hurwitz).** The shared calibrator first reweights the full universe to the targets, and records are then drawn with probability proportional to those weights, with replacement; each of the n draws receives an equal share W/n of the total weight W, and repeated draws accumulate on the same record. This is the unbiased Hansen–Hurwitz integerisation (Hansen and Hurwitz, 1943) of the dense weights, which preserves their expectation at every budget, and is the calibrate-then-reduce baseline.

This design does not claim that raking, GREG, balanced sampling, or combinatorial optimisation are irrelevant in general. They remain the reference methods for simpler margin surfaces, classical calibration, target-balanced survey designs, and discrete synthetic-population construction. The empirical comparison below focuses on the methods that can be applied to the full Populace target surface under the same scoring loss and that were run in the real-data sweep.

Holding the candidate universe and the target set fixed isolates the effect of how records are chosen. Every method starts from the same candidate frame with its weights reset to a uniform prior, so each record's initial weight $w_{0,i}$ is one shared constant. We sweep the retained-record budget across 2,000, 5,000, 10,000, 20,000, and 40,000 records, from aggressive compression of the candidate universe up to a large fraction of it, and run each budget over three calibration seeds (seeds 0–2). Informed L_0 sets the achieved budget at each grid point through its λ_{L_0} bisection, and the baselines are matched to the same retained count.

4.5 Out-of-sample target evaluation

A sampler can match the targets it is fit to and still generalize poorly. To test generalization, the design separates the targets used to fit the weights from the targets used to score the result, and holds out whole target families rather than a random subset of targets. The reason is structural: targets within a family nest (a national total is the sum of its state cells), so a random target split leaks: a held-out cell is nearly determined by its retained siblings, which would overstate generalization. Holding out a whole family removes that leakage.

The fixed holdout comprises the `cms_medicaid`, `usda_snap`, and `state_income_tax` families, 201 targets spanning three domains (Medicaid enrollment and spending, SNAP, and state income tax). These families are held out of every method's fit and scored only out-of-sample. Each held-out family keeps a related family in the fit, so the test asks each method to reproduce a held-out family while a structurally similar family remains in the fit, rather than orphaning a domain entirely (Table 3). The `cbo` family is additionally held out as a POPULACE validation-only family (forecast and aging projections used as diagnostics rather than contemporaneous targets), so it is never fit. Of the 4,393 targets in the system, 4,187 are used to fit the weights and 206 are held out (the 201 contemporaneous targets plus the 5 `cbo` validation-only targets).

Held-out family	Domain	Related family kept in the fit
`cms_medicaid`	Medicaid enrollment and spending	`cms_aca`, `cms_medicare` (other health-program enrollment)
`state_income_tax`	State income tax	`irs_soi` (federal income-tax aggregates)
`usda_snap`	SNAP enrollment and spending	`hhs_acf_tanf` (a related means-tested transfer)
`cbo`	CBO revenue and aging projections	none; validation-only, never fit

Table 3: The fixed family-level holdout. Each contemporaneous held-out family keeps a structurally related family in the fit, so generalization is tested against a similar but unseen family rather than an orphaned domain. The `cbo` family is held out as a validation-only diagnostic.

Family counts are read from the run manifest. The related-family pairing is a property of the target system, not of any method.

As a stricter robustness check we also rotate whole families through the holdout in a family-grouped five-fold design: the families are dealt into five folds balanced by target count, and each fold is held out in turn. Because the folds hold out structurally different families rather than independent draws, we report them per fold rather than as a single inferential test (Section 5.2); the rotation surfaces in particular the fold that holds out the dominant `irs_soi` family.

4.6 Evaluation metrics

Each run reports the metrics in Table 4 for the scored targets, both in-sample and out-of-sample. They fall into two groups: accuracy measures, which ask how closely the weighted estimates reproduce the targets, and representativeness measures, which ask at what cost in weight concentration and compute that accuracy is bought.

Metric	Definition	Why reported
Median ARE	Median absolute relative error $\lvert\hat{t}_j - t_j\rvert / \lvert t_j\rvert$ over scored targets	Headline accuracy; robust to the near-zero-denominator tail
Mean and max ARE	Mean and maximum of the same per-target errors	Reported alongside the median, flagged as tail-sensitive
Micro vs. macro average	Mean over all targets vs. mean of the per-family means	The micro mean is dominated by `irs_soi` (71% of targets); the macro mean weights each family equally
Effective sample size	$\mathrm{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$ and the implied design effect	A sampler can fit the targets while concentrating population mass on a few records, so ESS is a primary result, not a footnote
Retained count	Non-zero record count	Sample size
Max weight	Largest fitted weight	Weight concentration
Runtime	Wall-clock time	Compute cost

Table 4: Metrics reported for each run, in-sample and out-of-sample. Accuracy is led by the median ARE; the effective sample size measures the weight concentration that accuracy is achieved at.

Errors are reported by geographic level and by target family as well as in aggregate, because a sampler can match national totals while missing local ones, or match one source while missing another.

A small number of targets have denominators so small that a single record can swing their relative error past 100 per cent. We identify these structurally rather than by an arbitrary cutoff: target j is denominator-degenerate when its value is smaller than its identifiability floor $\max_i |M_{ji}| w_{0,i}$, the largest contribution any single record makes at its initial weight. For such a target the absolute relative error reflects integer placement noise rather than calibration quality. Rather than winsorize or trim the error distribution, we name these targets in a one-time audit and report a targeted-removal sensitivity (the mean recomputed with exactly those named targets dropped) beside the full mean, so the effect of the degenerate targets is visible and attributable. Target counts are read from each run's manifest.

5 Results

The results address the questions from the introduction: when informed L_0 selection reproduces the targets more accurately than the matched-budget baselines, whether the retained sample generalizes to held-out target families, and whether the size, geography, and concentration controls behave as intended. The reported runs share one frozen candidate universe and a fixed target system. The `cms_medicaid`, `usda_snap`, and `state_income_tax` families (and the `cbo` validation-only family) are held out of every method's fit and scored only out-of-sample (Section 4.5). Each budget is run over three seeds; cells report the cross-seed median or mean as noted, with a 95% confidence interval where shown. Accuracy is led by the median absolute relative error, with the mean reported alongside but treated as tail-sensitive (Section 4.6).

5.1 Informed versus uninformed sampling

Figure 2 is the central result: out-of-sample calibration error against the requested record budget, swept from aggressive compression up to the largest budget, for informed L_0 selection, informed L_1 selection, random sampling with reweighting, and the Hansen–Hurwitz survey-weight baseline. We plot against the requested budget because each method prunes to its own achieved count near that request—within three per cent for informed L_0 and L_1 (Section 5.3). The regimes are distinct, and the median and the mean tell different stories. Under aggressive compression informed L_0 holds out-of-sample error bounded while random + reweight cannot fit the target system from so small a draw, with survey-weight sampling between them; the contrast is starkest on the tail-sensitive mean, where at the 2,000-record budget informed L_0's 225% sits against random + reweight's 1,798% (Table 6). On the headline median the methods converge quickly: the edge passes to random + reweight from about 5,000 records up, so informed selection leads on the median only under the most aggressive compression. Informed L_0 keeps the lower mean at every budget, because the inflated mean tracks the seed-to-seed instability of a small random or survey draw that informed selection avoids. Informed L_1 is the weakest arm throughout: forcing sparsity through a single penalty collapses its fit, with its median pinned near 100% at the tightest budgets, and it converges toward informed L_0 only at the largest budget (Section 6). The frontier reads as graceful degradation under compression: informed L_0 is decisive where the budget is too small for a random draw to span the targets, and once the budget is large enough that almost any subset can be reweighted the median edge passes to random + reweight while informed L_0 keeps the lower tail. A looser loss cap sharpens that small-budget edge, making informed L_0 the most accurate on the median too at the most drastic budgets (Appendix B).

Table 5 reports the four methods at the anchor budget with the operational diagnostics, in particular the effective sample size that the matched accuracy is achieved at.

Method	Budget	In-sample median ARE (%)	Out-of-sample median ARE (%)	Out-of-sample mean ARE (%)	ESS	Max weight
Informed L_0	10,000	35.3	35.0	584.0	1,993	665,674
Informed L_1	10,000	667.9	356.9	5,724.7	2,370	296,974
Random + reweight	10,000	0.6	31.3	1,171.7	1,313	1,095,575
Survey-weight sampling	10,000	29.1	31.2	1,022.1	3,090	1,268,715

Table 5: Informed L_0 sampling against the matched-budget baselines at a representative budget (the $\sim$10,000-record anchor of Figure 2), at $\lambda_{L_2} = 0$. All methods share the candidate universe, the target set, and the fixed family-level holdout, and differ only in how records are selected. ARE is absolute relative error across the scored targets; the median is the headline read and the mean is reported alongside as tail-sensitive. ESS is the Kish effective sample size, reported as an output diagnostic because a sampler can fit the targets while concentrating population mass on a few records. Cells are the cross-seed summary over 3 seeds.

The matched record budget and target counts are read from each run's manifest; the out-of-sample columns score the held-out families. The survey-weight baseline is the unbiased Hansen–Hurwitz integerisation (probability-proportional-to-size with replacement). Informed L_0 and informed L_1 runtimes include the outer budget bisection that sets their penalty; the matched baselines run once at the budget. Runtime and the family macro-average are reported in the run summary.

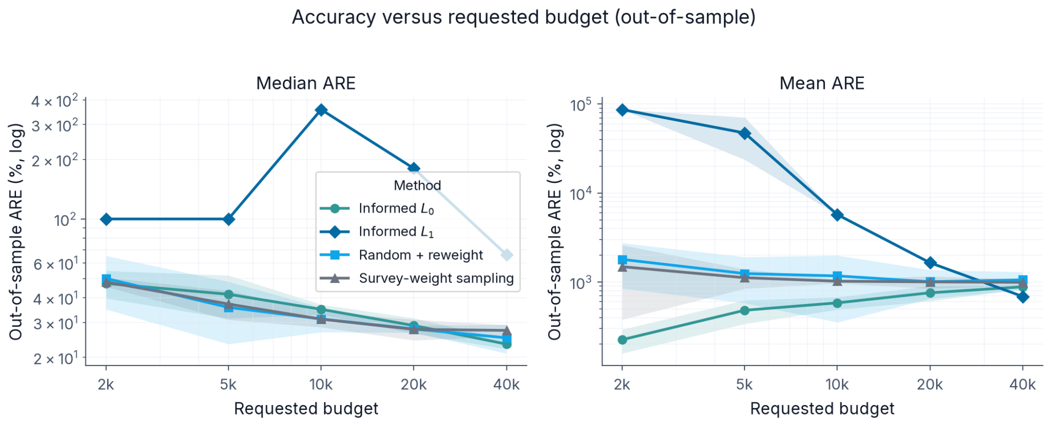

Figure 2: Out-of-sample calibration error against the requested record budget, for informed L_0 selection, informed L_1 selection, random sampling with reweighting, and the Hansen–Hurwitz survey-weight baseline. The mean and median panels are reported together because a few near-zero-denominator targets inflate the mean (Section 5.5); the median is less sensitive to that tail, and both panels are on a logarithmic scale. Lines are the mean over three seeds and bands the 95% confidence interval.

The hypothesis behind informed selection is that it lowers calibration error at a fixed budget, because the gates retain the records the targets constrain. Table 6 reports the same-seed difference against random + reweight at each budget, on the headline median and the tail-sensitive mean; with only three seeds, the paired t-statistics are descriptive diagnostics rather than strong inferential evidence. The two metrics split. On the median informed L_0 leads only under aggressive compression, and random + reweight takes over from about 5,000 records up ($p = 0.016$ at 5,000), the same crossover the frontier shows. On the mean informed L_0 leads at every budget, significantly so under aggressive compression ($p = 0.017$ at 2,000 and $p = 0.026$ at 5,000) and narrowing as the draws stabilise, because the tail is what random + reweight cannot control while on the typical (median) target the two draw level once the budget spans the system.

	Median ARE (%)			Mean ARE (%)		
Budget	Informed L_0	Random + reweight	Δ (pp)	Informed L_0	Random + reweight	Δ (pp)
2,000	47.3	50.1	-2.8 [-16.9, 11.3]	225.0	1,797.5	-1,572.5 [-2,464.6, -680.4]
5,000	41.7	35.8	5.9 [2.7, 9.1]	481.9	1,246.3	-764.4 [-1,304.9, -224.0]
10,000	35.0	31.3	3.7 [0.8, 6.6]	584.0	1,171.7	-587.7 [-1,380.3, 204.8]
20,000	29.0	27.8	1.1 [-1.0, 3.3]	757.7	1,009.0	-251.3 [-582.2, 79.6]
40,000	23.3	25.1	-1.8 [-7.2, 3.6]	884.3	1,059.0	-174.6 [-401.1, 51.8]

Table 6: Paired out-of-sample comparison of Informed L_0 against Random + reweight at each budget, on the headline median and the tail-sensitive mean. Each Δ is the cross-seed mean of the per-seed difference between the methods (negative favours Informed L_0) with a 95% confidence interval over the 3 paired seeds; a paired t-test on these differences is reported in the text as a descriptive diagnostic. On the median Informed L_0 leads only under aggressive compression, where a small random draw cannot span the targets; on the mean it leads at every budget, because the mean tracks the seed-to-seed tail that Random + reweight cannot control. Numbers are the un-penalised ($\lambda_{L_2} = 0$) rows.

5.2 Generalization to held-out target families

The out-of-sample columns above test generalization against the fixed family-level holdout: the `cms_medicaid`, `usda_snap`, and `state_income_tax` families are scored but never fit, so no method can have tuned to them, and each keeps a sibling family in the fit (Section 4.5). Figure 3 plots the gap between out-of-sample and in-sample error across the budget sweep, and Figure 4 breaks the out-of-sample error down by held-out family at the representative budget.

In-sample, informed L_0 fits the targets it sees no more tightly than it reproduces unseen ones: its in-sample and out-of-sample median ARE are essentially level (Table 5). Random + reweight instead fits the in-sample targets far more tightly—a median of 0.6% at 10,000 records—but that fit does not survive on families it never saw, where its median rises to about 31%, while informed L_0 carries almost no such gap. Informed L_0 spends the tight budget on records that generalize across families rather than on driving the in-sample fit down. Figure 3 shows the same ordering on the mean, where the near-zero-denominator tail (Section 5.5) enlarges every gap: informed L_0's mean gap stays near zero, random + reweight's runs to several hundred points, and informed L_1's swings widest. The held-out families differ in difficulty (Figure 4): on the two well-conditioned families, `cms_medicaid` and `usda_snap`, the methods are close, while the state income-tax (`census_stc`) and validation-only `cbo` families are dominated by near-zero-denominator targets—44 of the held-out targets are degenerate state income-tax cells (Section 5.5)—so every method's mean there runs into the thousands of per cent and the family comparison is uninformative. Informed L_1 is the worst arm on every family.

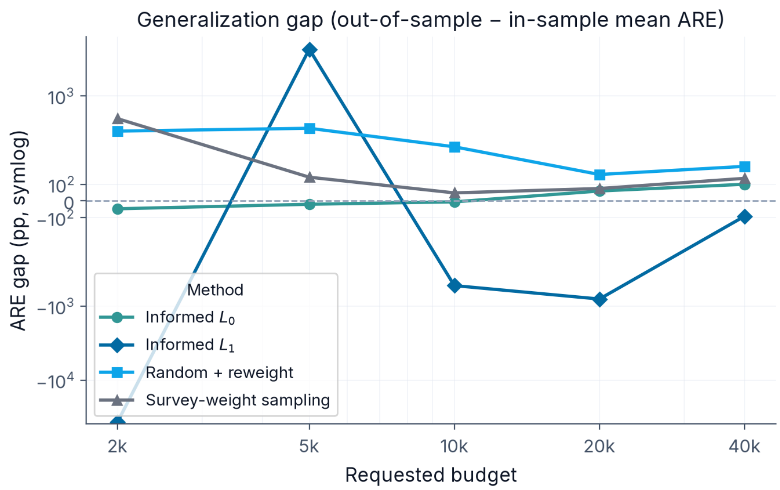

Figure 3: Generalization gap: out-of-sample minus in-sample mean ARE against the requested record budget, for the four methods, on a symmetric-log scale. A gap near zero means the retained sample reproduces the held-out families about as well as the families it was fit to; a positive gap measures the cost of generalization, and a negative gap marks a case where the in-sample mean is as inflated as the out-of-sample mean. Each line is the mean over three seeds.

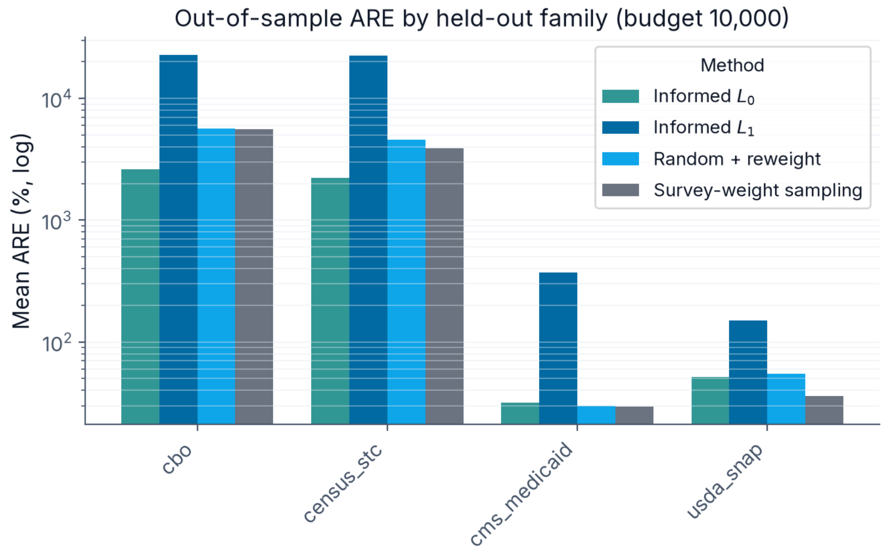

Figure 4: Out-of-sample mean ARE by held-out family at the ~10,000-record budget, for the four methods (fixed family-level holdout, $\lambda_{L_2} = 0$), on a logarithmic scale. `census_stc` is the state income-tax family and `cbo` is the validation-only family (Section 4.5). The per-family view shows which held-out families each method reproduces, beneath the aggregate frontier of Figure 2.

As a stricter, descriptive robustness check we rotate whole families through the holdout in a family-grouped five-fold design and report each fold separately rather than as a single inferential test. Informed L_0 has the lowest median out-of-sample ARE in two of the five folds, and random + reweight leads on the other three. On the target-weighted mean across folds the ranking favours the baselines: informed L_0 and informed L_1 average 2,635% and 2,650% against 1,299% for random + reweight and 1,440% for survey-weight. The gap comes from the fold that holds out `irs_soi`, the family supplying 71% of the targets. Removing it leaves a thin, badly conditioned fit surface on which every method's mean is dominated by a few near-zero-denominator targets, and informed L_0's mean is highest there (3,658% against 1,745% for random + reweight). The medians on that same fold are close (89% for informed L_0, 79% for random + reweight, 83% for survey-weight), which places the gap in the tail rather than the typical target. We report the rotation per fold and treat the `irs_soi` fold as a limitation rather than a headline comparison.

5.3 The size and concentration controls

A second question is whether the framework gives usable control over the output. The size control works as designed: λ_{L_0} sets the retained-record count, and the eight-step bisection over λ_{L_0} hits each requested budget to within about three per cent (Appendix C). Because the sparsity penalty produces exact zeros, the dataset size is an output of the optimisation rather than a post-processing cutoff.

Weight concentration is a measured trade-off rather than a fixed setting. The two concentration controls act on the ratio of each fitted weight to its initial weight (Section 4): the soft L_2 penalty shapes the whole distribution, and the hard `max_weight_ratio` caps the largest weight. Figure 5 reports the effective sample size and the largest fitted weight across methods and budgets, and Figure 6 compares informed L_0 with and without the L_2 penalty ($\lambda_{L_2} = 0$ against $\lambda_{L_2} = 10^{-4}$) across the budget sweep, tracing what raising the effective sample size costs in out-of-sample accuracy. The operability picture is budget-dependent (Figure 6). At the larger budgets the L_2 penalty buys effective sample size cheaply: at 40,000 records $\lambda_{L_2} = 10^{-4}$ raises the effective sample size by about 70% (from 3,803 to 6,482) for under two points of median out-of-sample ARE. At the smallest budget the same penalty is costly on the tail, inflating the 2,000-record out-of-sample mean ARE because there the fit already depends on concentrating weight onto a few records. We report the un-penalised frontier ($\lambda_{L_2} = 0$) as the accuracy result and treat $\lambda_{L_2} = 10^{-4}$ as an operability setting: it earns its accuracy cost where the budget is large, not where it is tight.

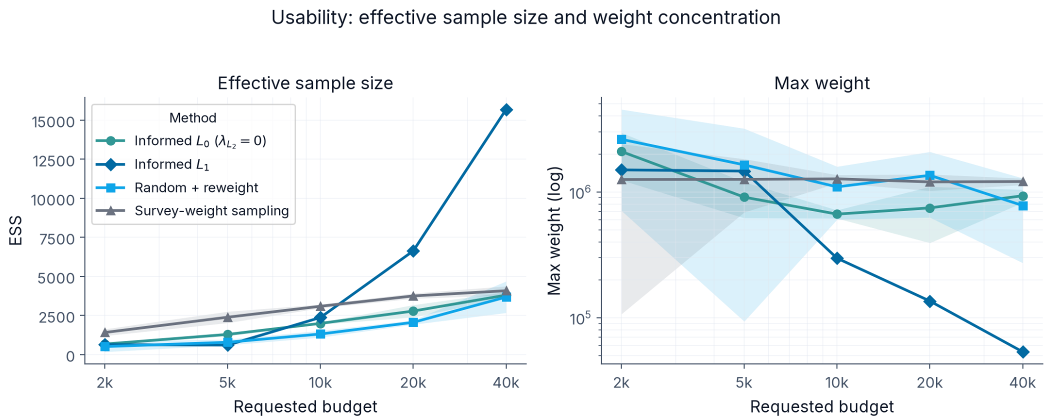

Figure 5: Effective sample size (left) and the largest fitted weight (right) against the requested record budget, for the four methods at $\lambda_{L_2} = 0$. An effective sample size well below the retained-record count signals weight concentration: a sampler can fit the targets while loading population mass onto a few records. Lines are the mean over three seeds, bands the 95% confidence interval.

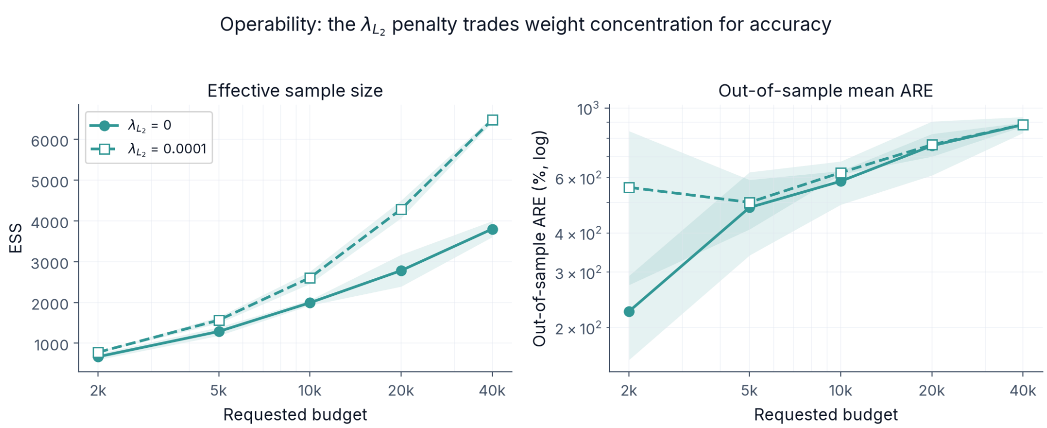

Figure 6: Operability of the λ_{L_2} penalty for informed L_0, across the budget sweep: the effective sample size it buys (left) against the out-of-sample mean ARE it costs (right), with $\lambda_{L_2} = 0$ (solid) and $\lambda_{L_2} = 10^{-4}$ (dashed). Lines are the mean over three seeds, bands the 95% confidence interval.

5.4 Accuracy by geography

Table 7 reports absolute relative error by geographic level for the informed L_0 run, since a sampler can match national totals while leaving local totals off. The current target surface is national and state only; no congressional-district targets are in the prioritized subset, so that level is not scored.

Geographic level	Targets	Median ARE (%)	Mean ARE (%)	Max ARE (%)
National	129	31.3	458.6	9,777.2
State	4,058	35.3	593.6	81,987.9
All scored targets	4,187	35.3	589.5	81,987.9

Table 7: Calibration accuracy by geographic level for the informed L_0 run at the representative $\sim$10,000-record budget, measured as absolute relative error (ARE) across the in-sample targets. The median reports the typical target's error; the mean and maximum are inflated by a handful of near-zero-denominator targets whose relative error is large despite small absolute misses (Section 5.5).

Target counts are read from the run manifest. No congressional-district targets are in the prioritized subset, so that level is not scored. Out-of-sample geographic accuracy is not broken out here; the aggregate held-out error appears in Table 5 and Figure 2.

5.5 Degenerate targets and metric robustness

A small number of targets are denominator-degenerate: their value falls below their identifiability floor, so a single record's placement swings their relative error past 100 per cent (Section 4.6). In the fit set, 741 of the 4,187 scored targets are degenerate by this test, and 735 of them are `irs_soi` cells (chiefly the state EITC-by-number-of-children counts); the remaining six fall in `hhs_acf_tanf` and `cms_aca`. The 44 degenerate targets in the held-out set are state income tax cells (`census_stc`) and one `cbo` target. We name these targets in a one-time audit rather than winsorize or trim, and report the mean recomputed with exactly them dropped beside the full mean. At the 10,000-record budget this targeted removal lowers informed L_0's in-sample mean ARE from 589% to 64%, against a median of 35%; the degenerate targets barely move the median, which is why we lead with it.

6 Discussion

6.1 What the sampling comparison establishes

The comparison against the matched-budget baselines turns on the record budget, and on which error statistic is read. Under aggressive compression, where the budget is too small for a random draw to span the target system, informed L_0 selection holds the lowest error because the gates spend the budget on the records the targets constrain. As the budget grows the methods converge: on the headline median the edge passes to random + reweight once almost any subset can be reweighted to fit (near 5,000 records here), while on the tail-sensitive mean informed L_0 stays ahead at every budget, because it removes the seed-to-seed instability that swings a small random or survey draw's mean by thousands of per cent (Section 5). The result is graceful degradation under compression rather than a blanket accuracy win, and the budget at which the median crossover happens is itself a finding. Past it the case for informed L_0 rests on its controls, on its near-zero generalization gap, and on the tail it keeps bounded, rather than on a median accuracy gap. It removes the risk of a poor random draw and sets the dataset size, the geographic emphasis, and the weight concentration through explicit parameters in one optimization. That control matters most in the build-big-then-prune setting the method is designed for, where a pipeline assembles a candidate universe far larger than it can ship and must reduce it to a fixed budget under subnational calibration targets; this study tests the national and state surface, and the subnational case is left to future work. Reporting out-of-sample error on held-out families keeps the comparison honest, since any method matches the targets it was fit to more tightly than ones it never saw.

The convex L_1 arm sharpens the contrast. L_1 pursues the same target-informed sparse selection, but through a single penalty that must both decide which records to drop and how much to shrink the survivors. Forcing roughly two thousand of seventy-five thousand records to survive demands a large penalty, which soft-thresholds the retained weights toward zero and collapses the calibration fit: its out-of-sample median ARE is pinned near 100% at the tightest budgets and its mean runs to several tens of thousands of per cent, recovering toward informed L_0 only once the budget is loose enough that little shrinkage is needed (Figure 2). Informed L_0 avoids this because the Hard Concrete gate and the fitted weight are separate variables: the gate decides membership and the weight is free to take whatever positive value the targets require. Decoupling selection from shrinkage is what lets L_0 degrade gracefully where the convex relaxation does not, and it is the core of the L_0-versus-L_1 comparison. L_1 is more stable to optimize and can be the better choice where sparsity is not forced hard; the case for L_0 here rests on the joint selection-and-weighting framing and the explicit control over the retained count, not on a claim that it is the most accurate sparse method in every setting.

6.2 What the controls establish

The second contribution is operational: the same framework produces datasets of different sizes by changing the L_0 penalty, focuses on a geographic level by weighting its targets, and shapes the weight distribution through two concentration controls, a soft L_2 penalty and a hard per-record cap. We treat the last of these as a measured frontier rather than a fixed setting: sweeping the L_2 penalty traces what tightening concentration (raising the effective sample size and lowering the largest weights) costs in calibration accuracy, so an operator can choose a point on that trade-off rather than accept whatever concentration the unconstrained fit produces. A random-sampling baseline can also produce a dataset of a chosen size, but it cannot make the choice of records respond to the targets, and it controls concentration only through the reweighting step. The value here is that size, geographic focus, and weight stability are set through explicit parameters in one optimization rather than through separate heuristics.

6.3 Trade-offs and limitations

The method does not remove modelling judgment, and several choices stay consequential; the optimizer's hyperparameters are one such choice, reported in full in Appendix D.

6.3.1 Target selection

The target set is a prioritized subset, not a complete inventory. The reasons for including each family are stated in Appendix A, and the empirical claims are tied to the exact targets in each run's manifest. A different target set could change the comparison, and the paper does not claim the chosen set is maximal.

6.3.2 When informed selection helps

Informed selection should help most when the targets are specific and local, so the choice of records matters. When the targets are few and broad, a random draw may already contain the records needed to match them, and the advantage of informed selection narrows. The paper reports the regimes it tests and does not extrapolate beyond them.

6.3.3 Weight concentration and effective sample size

Matching a demanding fiscal target system from a uniform prior can require the optimizer to load a large share of the population onto a small number of records, so the effective sample size can fall well below the retained-record count even when the calibration fit is good. That concentration is acceptable for the aggregates the dataset is calibrated to, but it is a real cost for any downstream estimand that is sensitive to the weight distribution. We expose the cost rather than hide it: the effective sample size and the largest weights are reported as primary results, and the L_2 sweep shows how much accuracy raising the effective sample size costs. That cost depends on the budget. At 40,000 records $\lambda_{L_2} = 10^{-4}$ raises the effective sample size by about seventy per cent (from 3,803 to 6,482) for under two points of median accuracy, while at the tightest budget it inflates the error tail instead. The penalty is worth applying at large budgets and not at tight ones, which is why we report the unpenalised frontier as the accuracy result and the penalty as a separate operability setting.

6.3.4 Generalization across target families

Holding out whole families is a deliberately hard test of generalization, harder than holding out a random subset of targets, and it need not flatter any method. On the fixed holdout, informed L_0 generalizes with almost no gap between its in-sample and out-of-sample median error, while random + reweight fits the in-sample targets far more tightly but pays about a thirty-point gap on families it never saw; informed selection trades in-sample fit for a sample that carries over. The stricter family rotation is less forgiving. On the target-weighted mean across folds the ranking favours the baselines, but the gap is confined to the fold that holds out the dominant `irs_soi` family: with 71% of the targets removed, every method's mean is dominated by a few near-zero-denominator targets, and the fold medians stay close. We read the rotation as a limitation to be reported per fold rather than as a single headline number, and widening the held-out surface further is left to future work.

6.3.5 Degenerate targets and the choice of error metric

A handful of targets have denominators small enough that one record's placement swings their relative error past 100 per cent, which inflates the mean absolute relative error without reflecting calibration quality. Rather than winsorize or trim the distribution by an arbitrary cutoff, we lead with the median, name the affected targets in a one-time audit, and report the mean recomputed with exactly those named targets removed. This keeps the metric honest and the dropped targets visible and attributable, but it does mean the headline accuracy figure depends on a stated, auditable choice rather than on the raw mean alone.

6.3.6 Computational cost

Building the calibration matrix and running gradient-based optimization cost more than a small classical calibration, and informed L_0 adds the eight-step budget bisection on top of a single fit; Appendix G reports the runtime each method reaches against the accuracy it buys. That cost is justified only when the problem requires it, which is why the comparison against a simpler random-sampling baseline matters.

6.4 Generalizability

The data engineering here is specific to the United States, but the question is general. Many microsimulation projects combine sources into a candidate dataset larger than they can ship, and then have to reduce it. Where the reduction has to respect calibration targets and keep weights usable, framing it as informed sampling, rather than as a random draw followed by reweighting, may carry over to other countries and other engines. POPULACE makes that portability concrete, since the method is a configuration choice rather than a fixed pipeline.

7 Future work

This study is a proof of concept on POPULACE's national and state target surface. The production motivation for informed selection is subnational: calibrating one dataset to congressional-district totals while keeping district sums consistent with state and national totals. Reaching that surface means pairing a much larger candidate universe—records replicated across the geographic placements district targets require—with the finer target system the pipeline can already construct (Appendix A), and testing whether informed L_0 holds its small-budget edge when the targets are local and numerous. That setting also invites the head-to-head benchmark this paper sets up but does not run: against generalized regression and iterative proportional fitting (Section 2), the classical calibrators the sampler is expected to respect. Tanton et al. (2014) compare generalized regression with combinatorial optimisation for small-area reweighting, and a subnational build would place informed L_0 on the same footing.

A second thread is the build-large-then-prune regime the controls are designed for, in which a pipeline assembles a candidate universe far larger than it can ship—millions of records rather than the 75,112 used here—and prunes it to a deployable budget under the calibration targets. The compression ratios would be far more aggressive than the range tested here, which is where informed selection is expected to help most: a random draw from a tight budget is least likely to span a demanding target system, so the gates have the most to add. Whether their small-budget edge scales to that regime, and at what compute cost (Appendix G), is what that setting tests.

8 Conclusion

This paper studies L_0 regularization as a way to subsample a large microsimulation dataset down to a usable size. The problem arises because a pipeline built for fidelity combines many sources and adds record-level variation, which produces more candidate records than a deployable dataset can hold. The reduction to a fixed budget is a sampling choice, and the paper treats it as one.

The method uses Hard Concrete gates to select records and fit their weights in one optimization, with selection driven by the calibration targets. That makes the retained-record count an explicit control and keeps the reduced dataset calibrated to its targets. We compare the method against three matched-budget alternatives that share its calibrator—its convex L_1 analog, random sampling with reweighting, and survey-weight (Hansen–Hurwitz) sampling—and report how the size, geography, and weight controls behave.

Across the budget sweep, informed L_0 reaches lower out-of-sample error than random sampling under aggressive compression; the two converge as the budget grows, and on the median random sampling becomes the more accurate of them, while on the tail-sensitive mean informed L_0 stays ahead throughout. Survey-weight sampling is the least accurate of the classical baselines, and the convex L_1 analog collapses under forced sparsity, recovering toward informed L_0 only at large budgets. Informed selection's distinctive value is therefore operational: it sets the dataset size, the geographic emphasis, and the weight concentration through explicit parameters in one optimization, and it avoids the variance of a random draw at tight budgets.

The contribution is therefore twofold: a target-informed sampling method, and an open pipeline, POPULACE, that makes the method a configuration choice other projects can reuse. Section 7 sets out the subnational and build-large-then-prune extensions the method is designed for, as well as the raking, balanced-sampling, and combinatorial-optimisation comparators that become appropriate on target surfaces where their assumptions fit cleanly.

Conflict of interest

All authors are affiliated with PolicyEngine, the nonprofit organization that develops POPULACE and the l0-python package evaluated in this paper. The authors have no other competing interests.

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Data and code availability

The paper and experiment code are at https://github.com/PolicyEngine/l0-paper. The construction and calibration pipeline is implemented in POPULACE, PolicyEngine's microsimulation data stack, open source at https://github.com/PolicyEngine/populace; earlier prototype code in the archived microplex and microplex-us repositories is retained only as a migration reference. The administrative targets are source-backed facts from PolicyEngine Ledger, open source at https://github.com/PolicyEngine/arch-data. The l0-python PyTorch package that implements the Hard Concrete optimizer is available at https://github.com/PolicyEngine/l0-python. The candidate universe used here is the public Hugging Face dataset policyengine/populace-us, snapshot be80a14f, file populace_us_2024.h5, with H5 SHA-256 beginning f0af2519. The current reproduction workflow relies on public code and generated data artifacts.

Use of generative AI

The authors used Claude (Anthropic) and Codex (OpenAI) for code assistance and manuscript drafting. All technical content, methodology, and analysis were designed and validated by the authors.

A Calibration targets

This study calibrates to a prioritized subset of the targets the pipeline can construct. Every target is a source-backed fact from PolicyEngine Ledger (PolicyEngine, 2026a), the arch-data fact store that holds the source publications and preserves the provenance of each published value; Populace composes calibration targets from those facts. The subset here is a surface of the most important national and state aggregates across the major tax and transfer families (Table 8), closely following the families Populace calibrates in its production US build. We defer congressional-district and finer-geography targets to the full build-then-prune configuration that pairs a much larger candidate household count with the finer geographic surface; at this proof-of-concept household count (75,112), the national and state surface is the appropriate scope. The subset is reported here in full, so the empirical claims can be read against the exact target system used rather than against a pipeline-wide inventory, and it is not presented as a complete or maximal target set.

Table 8 lists each family with its administrative source. The system holds 4,393 targets across eleven families and two geographic levels (national and state). The `irs_soi` family supplies 3,107 of them (71%), so the micro-average over all targets is dominated by IRS income-tax aggregates; the family macro-average (Section 4.6) is the counterweight that gives each family equal weight.

Family	Administrative source	Levels	Targets
irs_soi	IRS Statistics of Income: Historic Table 2 state totals and congressional-district tables, aggregated to state and national	National, state	3,107
census_population	Census Bureau annual state resident population estimates	National, state	936
cms_medicaid	CMS Medicaid and CHIP applications, eligibility, and enrollment reports	National, state	104
cms_aca	CMS open-enrollment-period state-level public use file	State	102
usda_snap	USDA FNS SNAP monthly state participation and benefit summaries	National, state	52
state_income_tax	Census Bureau State Tax Collections (STC), item T40 individual income taxes	State	45
hhs_acf_tanf	HHS ACF federal TANF and state MOE financial data	National, state	30
ssa	SSA Annual Statistical Supplement, OASDI beneficiaries and benefits	National	6
cbo	CBO revenue projections by category (validation-only)	National	5
jct	JCT estimates of federal tax expenditures	National	5
cms_medicare	CMS Medicare Trustees Report	National	1
Total			**4,393**

Table 8: Calibration target families and their administrative sources. Counts and sources are read from the target registry (POPULACE build 18497a03ccd1). The cbo family is held out as a validation-only diagnostic and is never fit (Section 4.5).

B Calibration loss cap: $c = 1$ versus $c = 10$

The calibration loss caps each target's contribution at c (Equation 9), and above the cap the target's gradient is zero. The cap is therefore more than a reporting choice: it shapes the fit, as a target already missed by more than c contributes a constant and stops pulling the optimizer. The runs in the main text use Populace's production value $c = 1$. Relaxing it to the generic solver default $c = 10$ keeps targets active until they are missed by an order of magnitude, which changes informed L_0's behaviour most where the budget is tightest and most targets start far from their value.

Table 9 compares informed L_0 (out-of-sample, $\lambda_{L_2} = 0$) at the two caps. Under the looser cap informed L_0 is uniformly more accurate, and the gain is largest under aggressive compression: at the 2,000-record budget its out-of-sample median ARE falls from 47.3% to 39.8% and its mean from 225% to 54%. The median improvement is enough to change the ranking at the most drastic budgets: at $c = 10$ informed L_0's 2,000-record median of 39.8% is the best of any method (random + reweight is 57.0% and survey-weight 46.8% at that cap), whereas at the production $c = 1$ the three are level there (Section 5). The mechanism is the starved gradient: at a tight budget most targets sit far above $c = 1$, so the optimizer fits only the already-near targets and the tail inflates, while $c = 10$ leaves a richer gradient and a better-conditioned fit. We report the production $c = 1$ as the headline because it is what the deployed build uses; the $c = 10$ contrast shows that informed L_0's advantage at drastic pruning widens, not narrows, under an easier loss.

	Median ARE (%)		Mean ARE (%)	
Requested records	$c = 1$	$c = 10$	$c = 1$	$c = 10$
2,000	47.3	39.8	225.0	53.5
5,000	41.7	26.3	481.9	38.2
10,000	35.0	23.6	584.0	36.0
20,000	29.0	20.7	757.7	32.5
40,000	23.3	18.0	884.3	31.3

Table 9: Informed L_0 out-of-sample ARE at the production loss cap $c = 1$ (main text) versus the looser generic default $c = 10$, at $\lambda_{L_2} = 0$, as cross-seed means over three seeds. The two runs are identical in every other respect—candidate universe, target weighting, optimizer settings, and seeds—so the cap is the only difference. The looser cap keeps more targets contributing gradient and improves informed L_0 most under aggressive compression.

C From a record budget to the sparsity penalty

The sparsity penalty λ_{L_0} sets the retained-record count, but the map from a penalty to a count is monotone and problem-dependent rather than closed-form: a larger λ_{L_0} prunes more records, and how many depends on the candidate universe and the targets. Each build therefore specifies a requested record count, and an outer bisection adjusts λ_{L_0} over eight steps (each a full optimization) until the achieved count matches. Table 10 reports the λ_{L_0} that each requested count converged to (informed L_0, $\lambda_{L_2} = 0$), with the achieved count. The penalty spans about an order of magnitude across the budget range, and the achieved count lands within three per cent of the request at every point.

Requested records	λ_{L_0} (converged)	Achieved records
2,000	2.8×10^{-5}	2,032
5,000	1.3×10^{-5}	4,973
10,000	7.5×10^{-6}	9,763
20,000	3.9×10^{-6}	20,237
40,000	1.9×10^{-6}	40,109

Table 10: The sparsity penalty λ_{L_0} that each requested record count converged to under the eight-step budget bisection (informed L_0, $\lambda_{L_2} = 0$), with the achieved retained-record count. Values are means over 3 seeds. A larger λ_{L_0} prunes more records; the bisection sets it per build rather than by hand.

λ_{L_0} and the achieved count are read from each run's metrics; the achieved count is within three per cent of the requested count at every budget.

The two informed methods reach a record budget by the same outer bisection but through different penalties, and that difference is what the L_0-versus-L_1 comparison turns on (Table 11). Informed L_1 replaces the Hard Concrete term with a convex penalty on the fitted weights, implemented as $\lambda_{L_1} \frac{1}{n} \sum_i w_i / w_{0,i}$, the mean weight ratio; this is the L_1 analog of the L_2 concentration penalty in Equation 10. Because the candidate weights are reset to a uniform prior (Section 4.4), the denominator $w_{0,i}$ is one shared constant, so the penalty is proportional to the plain L_1 norm $\sum_i w_i$ of Equation 7—the constant folds into λ_{L_1}—and soft-thresholding the ratio is soft-thresholding the weight, as in Section 2.6. Each proximal step takes a gradient step and then soft-thresholds the ratios, $r_i \leftarrow \max(r_i - \eta\, \lambda_{L_1}/n,\, 0)$ with $r_i = w_i / w_{0,i}$, so a record whose weight ratio falls below the threshold is set to exactly zero. The outer bisection moves λ_{L_1} over the same eight steps used for λ_{L_0}, and the achieved counts land near the request (1,859, 4,951, 9,709, 20,378, and 40,156 records for the 2,000- to 40,000-record budgets). The structural difference is that the single penalty λ_{L_1} controls both selection and shrinkage: the same threshold that zeroes a record's weight also pulls every surviving weight toward zero, so a tight budget forces a large λ_{L_1} that collapses the fit (Section 5). Informed L_0 keeps the two separate—the Hard Concrete gate decides membership while the fitted weight is free to take whatever positive value the targets require—so the bisection over λ_{L_0} moves the count without shrinking the survivors.

	Informed L_0	Informed L_1
Penalty added to $\mathcal{L}_{\mathrm{cal}}$	$\lambda_{L_0} \sum_i \bar{z}_i$ (expected open gates)	$\lambda_{L_1} \frac{1}{n} \sum_i w_i/w_{0,i}$ (mean weight ratio)
Quantity penalized	Gate activation probabilities $\bar{z}_i$	Fitted weight ratios $w_i/w_{0,i}$
Optimizer	Adam on weights and gates	Proximal gradient: Adam step, then soft-threshold
Route to an exact zero	Deterministic gate closes (Equation 6)	Soft-threshold $r_i \leftarrow \max(r_i - \eta\, \lambda_{L_1}/n,\, 0)$
Budget $\rightarrow$ penalty	Eight-step bisection on $\log_{10} \lambda_{L_0}$	Eight-step bisection on $\log_{10} \lambda_{L_1}$
Bisection bracket	$[10^{-7},\, 10^{1}]$	$[10^{-3},\, 10^{2}]$
Selection vs. shrinkage	Separate: the gate selects, the weight is free	Coupled: one penalty does both

Table 11: How the two informed methods map a requested record budget to a sparsity penalty. Both wrap the shared calibrator in the same eight-step outer bisection (Table 10) and differ in the penalty that the bisection tunes. The coupling in the last row—one λ_{L_1} setting both the retained count and the shrinkage of the survivors—is the property the L_0-versus-L_1 comparison isolates (Section 6). Source: the POPULACE proximal (L_1) and Hard Concrete (L_0) solvers.

D Optimization hyperparameters

Table 12 lists the hyperparameters of the L_0 optimization with their roles. The gate parameters follow the Hard Concrete construction of Louizos et al. (2018). The sparsity penalty λ_{L_0} and the epoch count are set per build to reach a target dataset size and are reported with each run.

Parameter	Value	Role
Gate temperature β	0.25	Sharpness of the 0/1 gate transition
Left stretch γ	-0.1	Enables exact-zero gates after clipping
Right stretch ζ	1.1	Enables exact-one gates after clipping
Initial keep probability	0.999	Records start nearly fully active
Weight jitter SD	0.05	Log-space noise on weights at initialization
Logit jitter SD	0.01	Log-space noise on gate logits at initialization
Learning rate	0.02	Adam optimizer step size
Epochs	1,500	Training iterations
Target loss cap c	1	Caps a single target's contribution to $\mathcal{L}_{\text{cal}}$; the runs reported here use Populace's production US-fiscal value $c = 1$ (Appendix B contrasts the generic default $c = 10$)
λ_{L_0}	2×10^{-6} to 5×10^{-5}	Sparsity penalty, set per build by bisection to the target size (Appendix C)
λ_{L_2}	0 (headline), 10^{-4} (operability)	Soft concentration penalty on $(w_i/w_{0,i})^2$ (mean over records); informed L_0 only
Budget bisection steps	8	Re-runs of the optimization that tune λ_{L_0} to the target size
`max_weight_ratio`	None (uncapped)	Hard per-record weight cap: $w_i \leq$ ratio $\times$ $w_{0,i}$, clamped each step

Table 12: Hyperparameters for the L_0 optimization. Gate parameters follow Louizos et al. (2018). Source: the `l0-python` Hard Concrete implementation used by POPULACE.

E Algorithm pseudocode

Algorithm 1 presents the L_0-regularized calibration procedure.

Algorithm 1 L_0-regularized calibration with Hard Concrete gates

Require: Calibration matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$, targets $\mathbf{t} \in \mathbb{R}^m$, initial weights $\mathbf{w}_0 \in \mathbb{R}^n$, per-target loss weights $\boldsymbol{\omega} \in \mathbb{R}^m$

Require: Hyperparameters: λ_{L_0}, λ_{L_2}, weight cap r (`max_weight_ratio`; ∞ if uncapped), β, γ, ζ, learning rate η, epochs E

Ensure: Calibrated sparse weight vector $\hat{\mathbf{w}} \in \mathbb{R}^n$

Initialize $\log w_i \leftarrow \log w_{0,i} + \mathcal{N}(0, 0.05^2)$ for all i

Initialize $\log \alpha_i \leftarrow \text{logit}(0.999) + \mathcal{N}(0, 0.01^2)$ for all i

Initialize Adam optimizer with parameters $\{\log w_i, \log \alpha_i\}$ and learning rate η

for epoch $= 1$ to E **do**

  Sample Hard Concrete gates (training):

  for $i = 1$ to n **do**

    $u_i \sim \text{Uniform}(\epsilon, 1-\epsilon)$

    $s_i \leftarrow \sigma\left(\frac{\log u_i - \log(1-u_i) + \log \alpha_i}{\beta}\right)$

    $\bar{s}_i \leftarrow s_i(\zeta - \gamma) + \gamma$

    $z_i \leftarrow \min(1, \max(0, \bar{s}_i))$

  end for

  $w_i^{\text{eff}} \leftarrow \exp(\log w_i) \cdot z_i$ for all i

  $\hat{t}_j \leftarrow \sum_i M_{ji} \cdot w_i^{\text{eff}}$ for all j

  $\mathcal{L}_{\text{cal}} \leftarrow \dfrac{\sum_{j=1}^m \omega_j \, \min\left(\left|\frac{\hat{t}_j - t_j}{s_j}\right|, c\right)}{\sum_{j=1}^m \omega_j}$

  $\mathcal{L}_{L_0} \leftarrow \sum_{i=1}^n \sigma\left(\log \alpha_i - \beta \log \frac{-\gamma}{\zeta}\right)$

  $\mathcal{L}_{L_2} \leftarrow \frac{1}{n} \sum_{i=1}^n \left(\frac{\exp(\log w_i)}{w_{0,i}}\right)^2$ ▷ soft concentration; informed L_0 only

  $\mathcal{L} \leftarrow \mathcal{L}_{\text{cal}} + \lambda_{L_0}\mathcal{L}_{L_0} + \lambda_{L_2}\mathcal{L}_{L_2}$

  Backpropagate $\nabla_{\log w, \log \alpha} \mathcal{L}$

  Adam step

  Clamp $\log w_i \leftarrow \min(\log w_i,\ \log(r \cdot w_{0,i}))$ for all i ▷ per-record cap; no-op if $r = \infty$

end for

Deterministic inference:

for $i = 1$ to n **do**

  $z_i^{\text{det}} \leftarrow \min(1, \max(0,\ \sigma(\log \alpha_i)(\zeta - \gamma) + \gamma))$

  $\hat{w}_i \leftarrow \exp(\log w_i) \cdot z_i^{\text{det}}$

end for

return $\hat{\mathbf{w}}$

F Dataset assembly details

After optimization the fitted weight vector is mapped back to record form and used to assemble the deployable dataset. The assembly step keeps the active records, reconstructs their person and tax-unit memberships, derives geography from the stored assignment, and recomputes geography-dependent quantities. Two details matter for fidelity. Geography is derived from the same assignment used to build the calibration matrix, so the assembled dataset matches the universe the optimizer saw. Take-up draws are regenerated with the same deterministic seeds used during matrix construction, so take-up-dependent targets stay consistent between the calibration problem and the assembled dataset.

G Cost diagnostics

Figure 7 reports the wall-clock runtime of each method against the out-of-sample median ARE it reaches, across the budget sweep, so the accuracy comparison can be read alongside its compute cost.

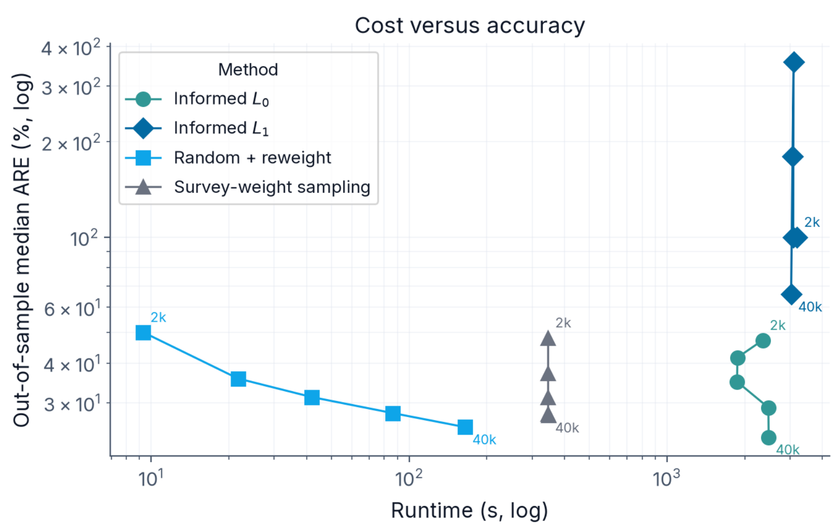

Figure 7: Cost against accuracy: wall-clock runtime (log scale) against out-of-sample median ARE (log scale) for the four methods, each point labelled by its requested record budget. Informed L_0 and informed L_1 runtimes include the eight-step budget bisection that sets their penalty; the matched baselines run once at the budget. Points are the mean over three seeds.

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